Foundation Model Insights and a Multi-Model Approach for Superior Fine-Grained One-shot Subset Selection

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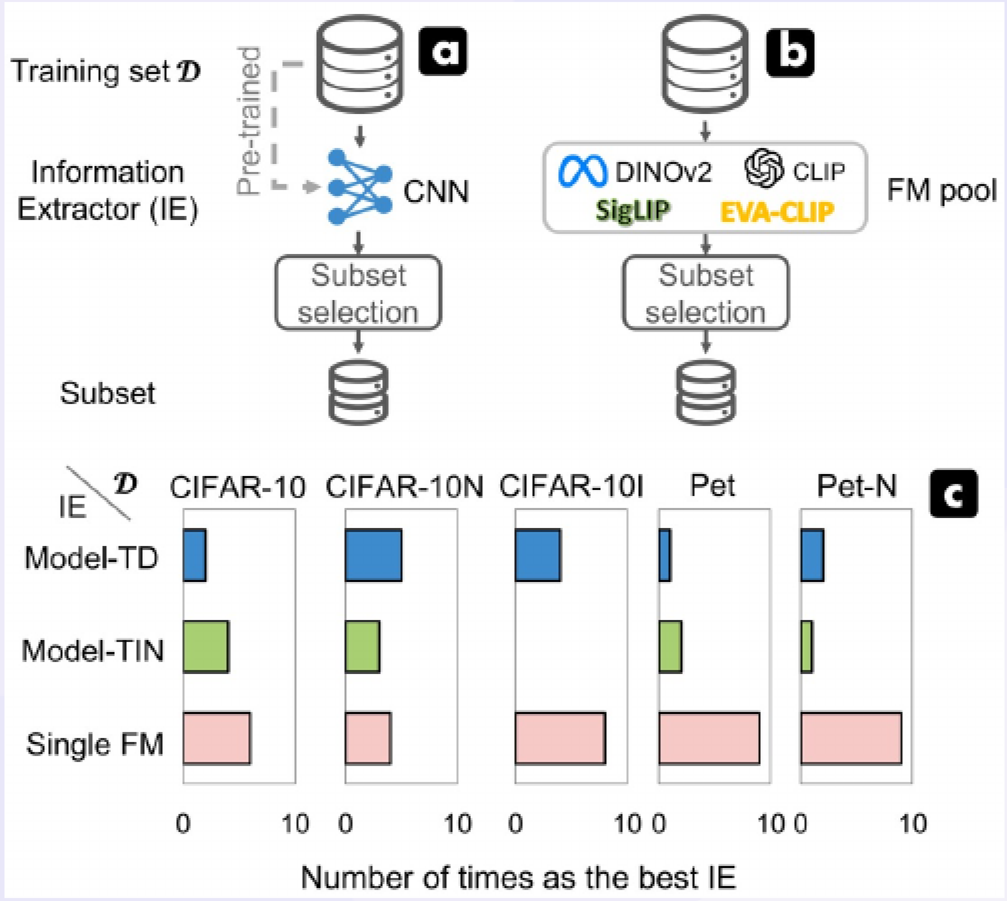
01 Framing the Problem and Headline Findings



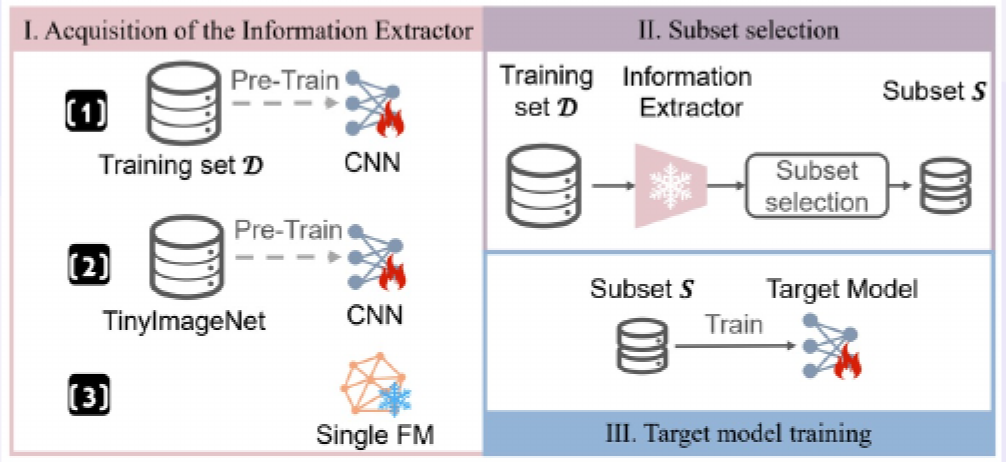
- One-shot subset selection requires a strong information extractor, while traditional extractors are dataset dependent and costly.
- Foundation models as extractors deliver consistent gains on fine-grained data and show reduced benefits on coarse or noisy data.
- RAM (Ranking Mean)–APL (Accuracy of Pseudo-class Labels), a multi–foundation-model selector combining intra- and inter-class signals, achieves **state-of-the-art** on Pet, Food-101, and CUB benchmarks.

02 Introduction and Contributions

- One-shot selection reduces training overhead by avoiding iterative rounds and eliminates target pretraining with foundation models.
- Experiments show dominant performance of foundation models on fine-grained datasets and weaker effects on noisy coarse datasets.
- RAM (Ranking Mean) + APL (Accuracy of Pseudo-class Labels) fuses CLIP and DINOv2 with intra- and inter-class signals to deliver **robust** state-of-the-art selection.

03 Related Work: One-shot Pipelines and FM-as-IE Gaps

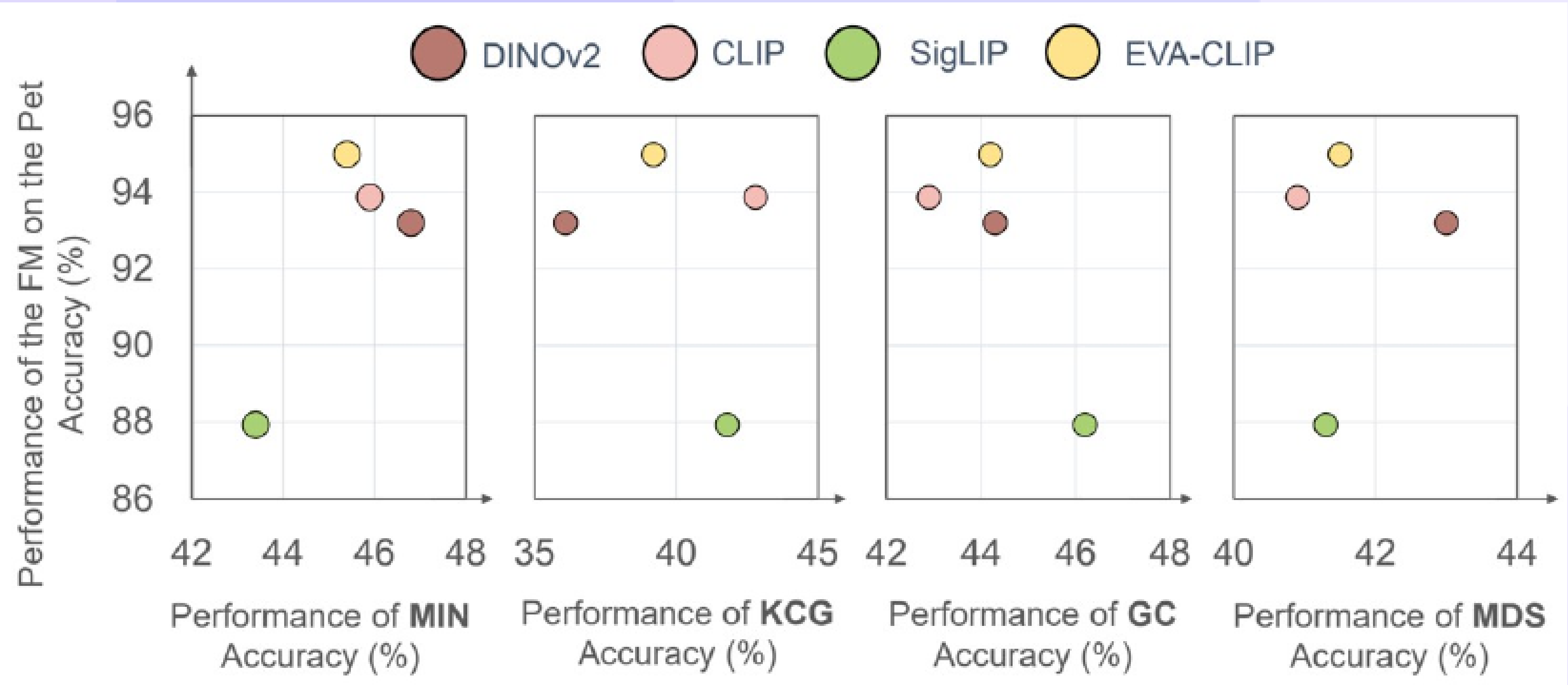
- One-shot selection couples an extractor, a measure, and a selector, while prior extractors rely on target pretraining.
- Foundation model extractors are promising yet show inconsistent dominance due to feature misalignment across models.
- Geometry methods favor diversity and boundary methods favor hardness, and fine-grained tasks require a **balanced** strategy.



04 Problem Setup: One-shot Subset Selection and the IE Role

- The goal is to select a budgeted subset that recovers full data performance with minimal loss.
- The choice of information extractor strongly influences selection quality and downstream accuracy.
- Foundation model extractors reduce dataset dependency and improve portability across domains.

05 Empirical Questions and Setup



- The study asks when foundation models are most effective and whether all foundation models perform equally well.
- Datasets include CIFAR variants and Pet (with a noisy version); extractors include target-trained models and multiple Foundation Models (FMs).
- Selectors include minimum distance (MIN), K-Center Greedy (KCG), Graph Cut (GC), and Moderate-Difficulty Sampling (MDS) under varied budgets.

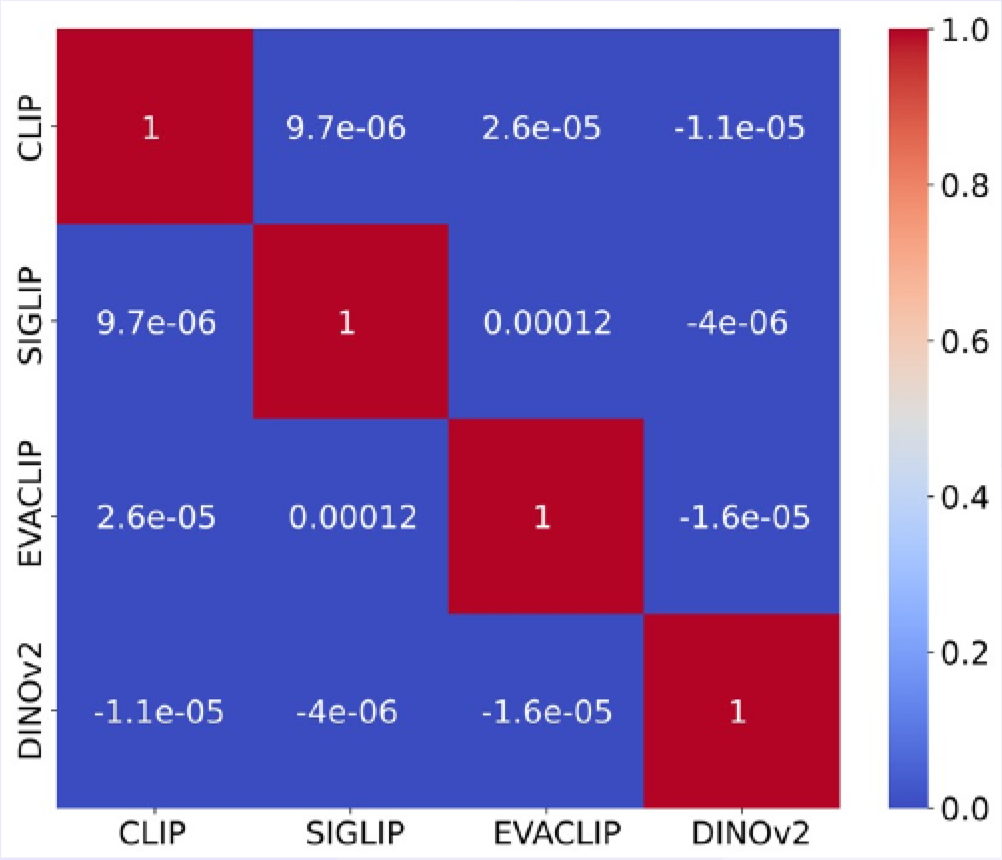
06 Key Observations and Implication: Motive for Multi–Foundation-Model (FM)

- Foundation models show strong advantages on fine-grained data and limited gains on coarse or noisy data.
- The optimal Foundation Model (FM) differs by dataset and budget, and downstream zero-shot accuracy does not predict selection quality.
- A multi model approach avoids per dataset tuning and stabilizes selection across conditions.

07 RAM-APL: Multi-FM Selection for Fine-Grained Data

- The method leverages multiple Foundation Models (FMs) and addresses unaligned feature spaces with **robust** scoring.
- The approach targets large within class variation and small between class margins characteristic of fine grained data.
- The design produces complementary gains over single model selection without extra tuning per dataset.

08 RAM-APL Components: RAM (Intra) + APL (Inter) and Scoring



- RAM (Ranking Mean) computes distance ranks to class centers across models to favor **representative** samples within each class.
- APL (Accuracy of Pseudo-class Labels) measures cross-model pseudo-label agreement to capture **hard** inter-class confusions.
- The final score blends RAM and APL with budget-aware weights and selects the lowest scoring samples.

09 Experimental Setup: Datasets, FMs, and Training

Dataset	Train	Test	Classes	Noise	Task
CIFAR-10	50,000	10,000	10	-	Single
CIFAR-10N-0.2Sym	50,000	10,000	10	20%	Single
CIFAR-10N	50,000	10,000	10	40.21%	Single
CIFAR-10I	20,431	10,000	10	-	Single
CIFAR-100	50,000	10,000	100	-	Single
Pet	3,680	3,669	37	-	Single&Multi
Pet-N	3,680	3,669	37	20%	Single
Pet-N-0.4Sym	3,680	3,669	37	40%	Single
CUB-200-2011	5,994	5,794	200	-	Single&Multi
Food-101	75,750	25,250	101	-	Multi

- The evaluation covers Pet, Food 101, and CUB along with additional CIFAR variants for breadth.
- CLIP and DINOv2 are used purely as feature extractors without any fine tuning for selection.
- The target models use standard training with stochastic gradient descent and release code for **reproducibility**.

10 Evaluation Protocol

- The main metric is target model test accuracy under class balanced sampling.
- Baselines are implemented through a unified framework with short target backbone pretraining (e.g., Confidence-based Active Learning (CAL), Graph Cut (GC)).
- Comparisons include classical selection strategies to ensure **fair** evaluation across budgets.
- Baseline sources: see Slide 'Baseline Methods and Citations' for full references.

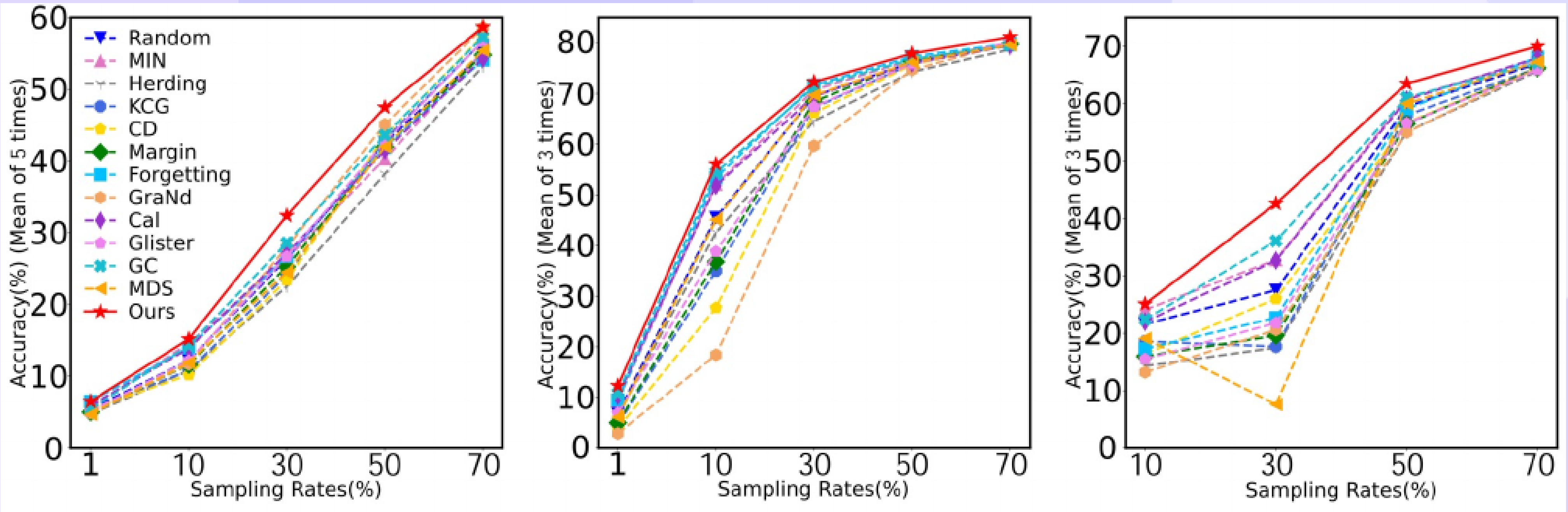
11 Baseline Methods and Citations

Methods	**IE**	**Sampling rates**					Average Improvement over Random
		1%	10%	30%	50%	70%	
Random	-	5.4±0.6	12.2±0.7	26.4±1.7	42.6±3.0	55.0±2.9	-
Herding		4.7±0.5	10.8±0.7	22.5±2.7	38.2±1.7	53.2±2.7	-2.44
KCG		5.4±0.6	10.8±0.9	24.1±3.0	43.0±2.9	56.7±2.2	-0.32
CD		5.4±0.6	10.2±0.9	23.4±3.3	43.5±2.0	57.5±2.5	-0.32
Margin		5.0±0.3	11.6±0.4	25.3±1.5	42.0±2.6	54.8±1.5	-0.58
Forgetting		6.5±0.7	14.0±1.6	26.7±1.5	41.9±0.9	54.1±3.0	+0.32
GraNd	Traditional	4.9±0.7	11.7±0.6	28.0±3.6	45.1±4.7	58.6±3.1	+1.34
Cal		5.9±0.6	13.9±0.8	27.1±1.4	41.5±1.2	54.3±3.7	+0.22
Glister		5.1±0.7	12.2±0.4	26.7±1.4	42.7±0.9	56.7±2.6	+0.36
GC		5.4±0.3	14.3±0.6	28.6±1.8	43.6±2.0	57.3±2.4	+1.52
MDS		4.7±0.4	11.9±1.6	24.5±2.3	42.0±2.2	55.4±3.4	-0.62
MIN		5.6±0.7	14.6±0.5	26.4±1.6	40.3±2.6	55.2±2.7	+0.10
Ours	CLIP+DINOv2	**6.5±0.4**	**15.2±1.2**	**32.4±2.9**	**47.5±1.9**	**58.7±2.2**	**+3.74**

- K-Center (core-set), Confidence-based Active Learning (CAL), Moderate-Difficulty (MD) sampling.
- The set further includes GraNd, Confidence based Active Learning, GLISTER, Graph Cut, Moderate Difficulty, and minimum distance.
- All baselines follow consistent one shot protocols to provide **reliable** head to head comparisons.

Rebuffi et al., 2017; Sener and Savarese, 2018; Birodkar et al., 2019; Tong and Koller, 2001; Toneva et al., 2019; Pa...

Main Results: RAM-APL vs. Baselines



(a) Oxford-IIIT Pet (b) Food-101 (c) CUB-200-2011

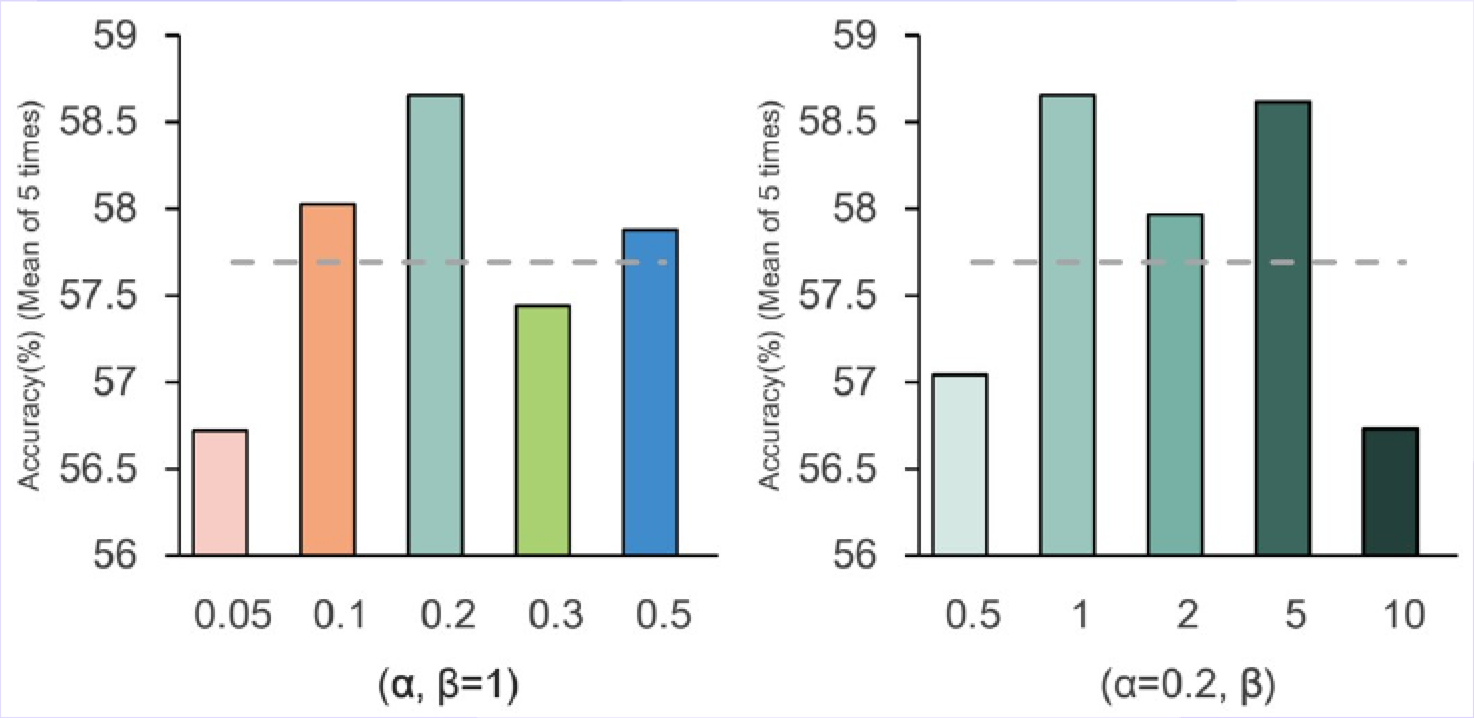
- RAM-APL (Ranking Mean + Accuracy of Pseudo-class Labels) consistently outperforms all baselines on Pet, Food-101, and CUB at every budget.
- The method yields notable gains over Random and surpasses strong submodular and geometric baselines.
- Average improvements are substantial on CUB and remain **stable** across architectures and seeds.

Ablations: RAM and APL Contributions

Method	**IE**	**Sampling rates**		
		1%	50%	70%
MIN	Model-TD	5.6±0.7	40.3±2.6	55.2±2.7
	CLIP	5.6±0.2	45.9±1.8	56.3±0.7
	DINOv2	6.2±0.1	46.8±2.0	**60.5±2.9**
RAM	CLIP+DINOv2	5.9±0.3	47.1±1.4	56.5±2.7
RAM-APL		**6.5±0.4**	**47.5±1.9**	58.7±2.2

- RAM with CLIP and DINOv2 surpasses minimum distance across budgets, confirming **effective** intra class fusion.
- RAM APL improves over single model selection at low and high budgets through better coverage and diversity.
- The combined score balances representative examples and confusing examples for strong generalization.

Analysis: Weights, Number of FMs, and Fusion



- The best configuration increases emphasis on inter class structure as the sampling budget grows.
- Multiple foundation models (FMs) outperform single models, and CLIP plus DINOv2 offers the best efficiency accuracy trade off.
- Simple feature concatenation underperforms principled fusion, especially at higher budgets with \noticeable drops.

Key Takeaways

- Foundation models as information extractors enable strong, portable one-shot selection on fine-grained data.
- Variability across single models motivates RAM (Ranking Mean; intra-class) + APL (Accuracy of Pseudo-class Labels; inter-class) fusion.
- State-of-the-art results on Oxford-IIIT Pet, Food-101, and CUB-200-2011.

Open Questions and Next Steps

- Adaptive weighting across datasets and budgets without labeled validation remains an **open** challenge.
- Extensions to detection, segmentation, and multimodal tasks can broaden applicability.
- Scaling to web scale and streaming settings and developing theory for guarantees are important next steps.

Final Takeaway

RAM-APL, a multi–foundation-model selector combining intra- and inter-class signals, delivers state-of-the-art one-shot subset selection on fine-grained datasets without target pretraining.

Thank You

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Questions & Discussion